

Raspberry Pi Plant Monitoring Kit:

Getting Started

Firstly, thank you for purchasing our plant monitoring and watering system – this is your gateway to a healthier and easier life.

Follow the below for a quick and easy setup of your system!

Getting to know the system

There are multiple parts to your plant monitoring system:

1. The plant sensors and watering equipment (the 'node'). There will be a solar panel attached to this.



2. The red box (the 'hub')



3. The USB-C cable for powering the hub



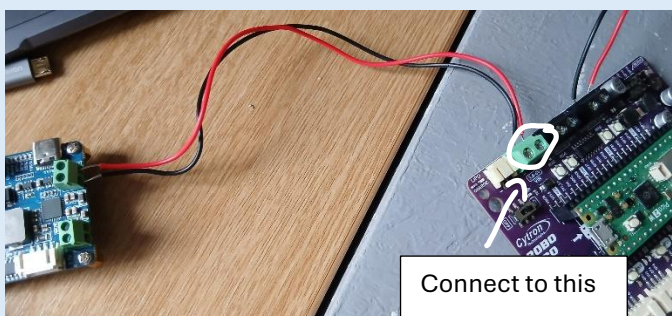
4. An Ethernet cable for connecting the hub to internet



Setup

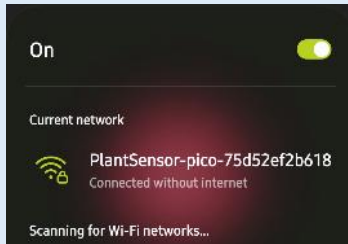
The system setup is very simple.

- 1) Connect the USB-C cable to the hub, and plug it into a wall socket
- 2) Connect the Ethernet cable to the hub and plug it into your Internet router
- 3) Connect the cables from the blue board to the green terminal on the node – you will need a screwdriver.



- 4) The sensor will now be broadcasting a Wi-Fi network, which will be called something like **PlantSensor-pico-xxxxxxxxxx** (the x will be a random number or letter). **Connect to**

this on your phone or computer. (If you're asked for a password, enter "*plant123*")



- 5) Open your web browser, and visit <http://192.168.4.1/>
- 6) You will be taken to a page where you need to enter your WiFi SSID and password.

Plant Sensor Setup

WiFi SSID:

Password:

Save

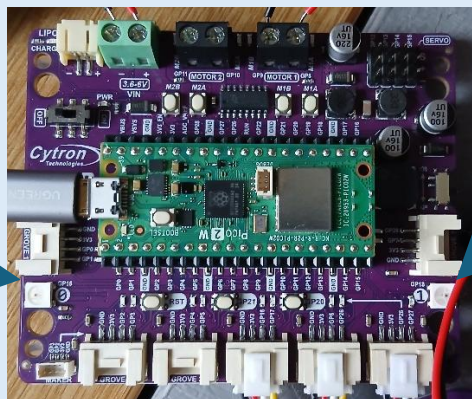
- 7) The sensor device will now restart, wait a few seconds
- 8) You will now be able to access the information dashboard by visiting <http://plantpi:5001/>
- 9) Your plant system will now work – enjoy your plants!

Using the Plant Monitor

The system will automatically water your plant if the moisture gets below the pre-defined level (around 20%). We found this works well for most plants, however in future releases we hope to be able to add a feature to let you change this.

The system should rarely need attention – however, if you get a mobile notification saying that the reservoir has run out of water, you will need to pay attention to this. This will also be shown through a red light on the left hand side light on the board.

This light will show the status. If it's green, everything's fine. If it's red, it means you need to fill up the reservoir!



This light will show the temperature. If it's red, the temperature is over 25 degrees. If it's yellow, the temperature is over 20 degrees.

If it's white, the temperature is over 15 degrees.

If it's blue, the temperature is under 15 degrees.

Setting up Mobile Notifications

This step is completely optional, and is only required if you want to stay up to date with your plant on the go.

- 1) On your phone, install the NTFY app.
- 2) Press the plus button
- 3) Enter a topic name (whatever you want)
- 4) Now go back to your plant dashboard. <http://plantpi:5001/>
- 5) Press the settings button, and enter <https://ntfy.sh/<yourtopic>> , replacing <your topic with what you set earlier
- 6) You will now be notified when the temperature reaches below 10°C, or above 25°C, or if the reservoir is out of water.

Troubleshooting

If your system doesn't seem to be working correctly, try turning the switch on the main board (purple) to off. Also unplug the hub. Wait a few seconds and plug the hub back in. After 2-3 minutes, turn the switch back to on.

If you still have issues, please contact us.